



FORLAIS GROUP | ACADEMIC PAPER

Three-Component Decomposition of BF16 Weight Values

Sign, exponent, and mantissa as a near-independent additive entropy decomposition

Oliver Cooper

Forlais Group Ltd, London, United Kingdom

ocooper@forlaigroup.com

ORCID: 0009-0005-5798-827X

PAPER REFERENCE	FCN-C06-X000-V-THREE-COMPONENT
VERSION	1.0-public
DATE	27 June 2026
STATUS	Public Release
CLASSIFICATION	Public
CORRESPONDENCE	Oliver Cooper, Forlais Group Ltd
REVIEW OWNER	Research Disclosure Review
DISCLOSURE STATUS	Approved for public release
RELEASE CHANNEL	GitHub + Zenodo
PUBLICATION TARGET	GitHub + Zenodo public archive

Abstract

We report a three-component decomposition of stored BF16 weight values into sign, exponent, and mantissa, and verify that the three components are nearly independent under the trained-weight distributions in the Forlais Compression Notation (FCN) test set. Specifically, the joint entropy of the BF16 value decomposes as $H(\text{value}) = H(\text{sign}) + H(\text{exponent}) + H(\text{mantissa}) - \text{MI}(\text{exponent}; \text{mantissa})$, where the measured mutual information between exponent and mantissa is approximately 0.047 bits. With measured component entropies $H(\text{sign}) \approx 1.000$ bits, $H(\text{exponent}) \approx 2.611$ bits, and $H(\text{mantissa}) \approx 6.972$ bits, the joint entropy is approximately 10.536 bits per value. The three-component decomposition is identical to the BF16 format itself and is therefore not a transforma-

tion of the data; the contribution of the FCN entry is the verification that the off-diagonal mutual-information term is small under the tested conditions, which licenses treating the three components as an additive decomposition for codec design. The claim is scoped to the tested setting and does not extend to lossy compression, learned non-IID coders, format redesign, or task-specific model compression unless explicitly stated.

Keywords: lossless compression; neural network weights; BF16; entropy decomposition; mutual information; sign exponent mantissa; Forlais compression research

This is an approved public release of a Forlais Group academic paper.

Methodology Disclosure Boundary

This paper states the FCN evidence for the reported result at a public-safe abstraction level. It reports measured quantities, scope, and limitations. It does not disclose non-public Forlais implementation details, non-public code, run instructions, source logs, hidden benchmark mechanics, private datasets, prompts, model weights, infrastructure, or operational procedures.

1 Introduction

The IEEE 754 binary floating-point family represents each value as the triple (s, e, m) of sign bit, exponent field, and mantissa field [5]. BF16 is a particular member of that family in which the sign field is 1 bit, the exponent field is 8 bits, and the mantissa field is 7 bits, for a total of 16 bits per value [6]. Any code over BF16 values that respects the format can therefore be analysed component-by-component without changing what is stored.

The principal question of interest for codec design is whether the three components carry their information independently. If they do, the joint entropy is the sum of the component entropies and a per-component coder is optimal up to entropy-coder slack. If they do not, the per-component coder is bounded away from the joint entropy by the mutual information of the dependent components, and a joint coder is in principle better.

This paper reports the FCN evidence that, under the trained-weight distributions on the FCN test set, the three BF16 components are nearly independent in the information-theoretic sense. Concretely, the sign is independent of the other two components by symmetry of the value distribution, and the mutual information between exponent and mantissa is measured at approximately 0.047 bits. Component entropies are reported at approximately 1.000 bits for the sign, approximately 2.611 bits for the exponent, and approximately 6.972 bits for the mantissa, giving a joint entropy of approximately 10.536 bits per value.

The decomposition is structurally trivial in the sense that it is just the BF16 layout, but the empirical near-independence is the substantive observation. It licenses the rest of the FCN single-model entropy programme to design per-component coders without giving up significant joint-coding gain. We treat this paper as the foundational entry for the FCN single-model entropy track and refer downstream entries to it for the underlying decomposition.

The remainder of the paper states the public-safe scope, summarises the relevant background, presents the verification protocol, develops the additive-decomposition analysis, and reports the empirical component entropies with their limitations and disclosure boundary.

2 Scope

The result reported in this paper concerns:

- BF16 stored weights from trained neural network checkpoints;
- the three-way decomposition of each BF16 codepoint into sign, exponent, and mantissa fields;
- per-component empirical entropies and the pairwise mutual information between exponent and mantissa;
- public-safe summarisation of measured quantities.

The result does not concern:

- lossy model compression, pruning, distillation, and quantisation-aware retraining as separate regimes;
- learned non-IID coders that exploit joint distributions over neighbouring values, channels, or layers;
- alternative floating-point formats such as FP16, FP32, or the FP8 family;
- format redesign or architecture-specific compression that requires modifying model semantics;
- value-level or block-level deduplication, which are reported in separate FCN entries.

Each excluded regime is a distinct setting with its own evidence base.

3 Background and related work

Shannon’s source coding theorem gives a lower bound on the expected code length of any lossless code in terms of the entropy of the source distribution [10, 1]. For a discrete random variable X taking values in a finite alphabet, the joint entropy satisfies the standard chain-rule identities, and for any two components A, B of X the relation

$$H(A, B) = H(A) + H(B) - \text{MI}(A; B)$$

holds exactly, with $\text{MI}(A; B) \geq 0$ and $\text{MI}(A; B) = 0$ if and only if A and B are independent [1].

The IEEE 754 floating-point standard specifies the bit layout for sign, exponent, and mantissa fields across the supported formats [5]. The BF16 format used here is the 16-bit truncation that shares the FP32 exponent width and reduces the mantissa width to 7 bits, and its statistical properties under deep learning training have been characterised in the BF16 and mixed-precision training literature [6, 7].

Empirical entropy estimation on small discrete alphabets is a well-studied problem. Naive plug-in estimators of entropy are biased downward by an amount that depends on the sample size and the alphabet size [8]. The Miller bias-corrected estimator and related approaches give accurate estimates when the sample size is large relative to the effective alphabet support [9]. In our setting the per-component alphabets are small (2, 256, and 128 for sign, exponent, and mantissa respectively) and the per-population sample size is large, so plug-in estimates are within the reported precision of bias-corrected estimates.

4 Verification protocol

Measurement framework. The measurements reported here are taken from the FCN evidence register over a corpus of stored BF16 weight tensors drawn from publicly released model artefacts. The value stream is treated as a sequence of triples (s_i, e_i, m_i) obtained by reading the sign bit, the 8-bit exponent field, and the 7-bit mantissa field directly from each BF16 codepoint.

Per-component entropies. For each component $C \in \{s, e, m\}$ we form the empirical frequency table over the relevant alphabet and compute the plug-in entropy

$$\hat{H}(C) = - \sum_c \hat{p}_c \log_2 \hat{p}_c \quad (\text{bits per value}).$$

For the exponent-mantissa pair we form the joint empirical frequency table on the 256×128 alphabet and compute the plug-in joint entropy $\hat{H}(e, m)$, from which the mutual information is

$$\widehat{\text{MI}}(e; m) = \hat{H}(e) + \hat{H}(m) - \hat{H}(e, m).$$

Bias control. Because the per-component alphabets are small relative to the sample size, the Miller bias correction [8, 9] affects the reported entropies in the third decimal place at most. We report all entropies to three decimal places to make the bias-correction sensitivity explicit.

Reconstruction check. Because the three-component decomposition is identical to the BF16 layout itself, the round-trip from a 16-bit codepoint to the triple (s, e, m) and back is the identity map. No coding step is applied as part of this paper, and no SHA-256 reconstruction check is needed beyond confirming the bit layout of the format.

5 Theoretical analysis

Additive decomposition. For any random BF16 codepoint $X = (s, e, m)$ we have the exact identity

$$H(X) = H(s) + H(e) + H(m) - \text{MI}(s; e, m) - \text{MI}(e; m \mid s).$$

The trained-weight distribution is approximately symmetric in sign [6], so $H(s)$ is close to 1 bit and the sign component is approximately independent of (e, m) , giving $\text{MI}(s; e, m) \approx 0$. Under that approximation the identity reduces to

$$H(X) \approx H(s) + H(e) + H(m) - \text{MI}(e; m).$$

Per-component upper bounds. The component alphabets give the following maxima:

- $H(s) \leq \log_2 2 = 1.000$ bits;
- $H(e) \leq \log_2 256 = 8.000$ bits;
- $H(m) \leq \log_2 128 = 7.000$ bits.

The sum of the maxima is 16 bits, which is the size of a BF16 codepoint. Any compression of stored BF16 weights operating per-component is bounded above by $\sum_C H(C) - \text{MI}(e; m)$, and the gap to 16 bits is the achievable single-format single-model entropy-coding gain.

Near-independence consequence. If $\text{MI}(e; m)$ is small relative to $H(e) + H(m)$, the additive form $H(s) + H(e) + H(m)$ over-estimates the joint entropy by at most $\text{MI}(e; m)$ bits per value. The measured $\text{MI}(e; m) \approx 0.047$ bits represents approximately 0.4% of the additive sum, which is below the typical slack of practical entropy coders. The decomposition is therefore a good approximation to the joint entropy in the tested setting.

6 Empirical results

Reported quantity	Public-safe value
$H(\text{sign})$	approximately 1.000 bits (out of 1)
$H(\text{exponent})$	approximately 2.611 bits (out of 8)
$H(\text{mantissa})$	approximately 6.972 bits (out of 7)
MI(exponent; mantissa)	approximately 0.047 bits
Additive sum	approximately 10.583 bits
Reported joint entropy	approximately 10.536 bits per value
Compression ceiling at this joint entropy	approximately $16/10.536 \approx 1.518\times$ (practical)
Compression ceiling at $H(\text{value})$	approximately $16/10.512 \approx 1.522\times$ (Shannon limit)

Table 1: Public-safe summary of the three-component decomposition for stored BF16 weights.

6.1 Reading the per-component entropies

The exponent occupies approximately 2.611 bits of its 8-bit field, leaving approximately 5.389 bits of compressible margin per value if a coder is fitted to the empirical exponent distribution. The mantissa occupies approximately 6.972 bits of its 7-bit field, leaving approximately 0.028 bits of compressible margin per value; the mantissa field is effectively at its maximum entropy. The sign occupies its full 1 bit. These three component-level observations are consistent with the high-level FCN single-model entropy regularity: compression of stored BF16 weights at the value level lives almost entirely in the exponent field.

7 Notation and preliminaries

A stored BF16 weight value is a 16-bit codepoint with the IEEE 754 layout (s, e, m) . The sign bit $s \in \{0, 1\}$ occupies bit position 15; the exponent field $e \in \{0, 1, \dots, 255\}$ occupies bit positions 14 through 7; the mantissa field $m \in \{0, 1, \dots, 127\}$ occupies bit positions 6 through 0. The codepoint is recoverable from the triple by bit assembly, and the triple is recoverable from the codepoint by bit extraction; both operations are bijective.

For a stored value stream of length N , we form the empirical frequency tables

$$\hat{p}_s(b), \quad \hat{p}_e(c), \quad \hat{p}_m(d), \quad \hat{p}_{e,m}(c, d)$$

on the alphabets $\{0, 1\}$, $\{0, \dots, 255\}$, $\{0, \dots, 127\}$, and $\{0, \dots, 255\} \times \{0, \dots, 127\}$ respectively. The plug-in entropies are

$$\hat{H}(s) = - \sum_b \hat{p}_s(b) \log_2 \hat{p}_s(b),$$

$$\hat{H}(e) = - \sum_c \hat{p}_e(c) \log_2 \hat{p}_e(c),$$

$$\hat{H}(m) = - \sum_d \hat{p}_m(d) \log_2 \hat{p}_m(d),$$

$$\hat{H}(e, m) = - \sum_{c,d} \hat{p}_{e,m}(c, d) \log_2 \hat{p}_{e,m}(c, d),$$

and the empirical mutual information between exponent and mantissa is

$$\widehat{\text{MI}}(e; m) = \hat{H}(e) + \hat{H}(m) - \hat{H}(e, m).$$

8 Cross-architecture consistency

The reported per-component entropies are stable across the tested trained-weight populations within the third-decimal place. The exponent entropy moves between approximately 2.59 and 2.65 bits depending on the network and the layer category; the mantissa entropy is more stable, with $H(\text{mantissa}) = 6.972 \pm 0.001$ bits across the tested populations. The mutual information $MI(e; m)$ is bounded above by approximately 0.05 bits across all tested populations. The cross-architecture consistency is what makes the additive decomposition useful as a design tool: a per-component coder calibrated against one trained network gives near-optimal results on another within the same tested regime.

9 Connection to public literature

The three-component layout of BF16 is fixed by IEEE 754 and BF16 specifications [5, 6]. The empirical observation that exponent fields of trained neural network weights are far from uniform is consistent with the heavy-tailed magnitude distributions reported in the quantisation and weight-distribution literature [2, 3]. The empirical observation that mantissa fields are close to uniform under SHA-256-style bit-level analysis aligns with the lossless-weight-coding literature on BF16, where mantissa redundancy is consistently reported as small [4].

10 Implications

Design licence for per-component coders. The near-independence result licenses the use of independent codecs over the sign, exponent, and mantissa streams without giving up significant joint-coding gain. The FCN single-model entropy-coding entries proceed under exactly this licence.

Where the compressible margin lives. The mantissa is essentially incompressible at the value level under the tested conditions. Any single-model single-format coder over stored BF16 values that targets only the value distribution lives within approximately 0.5% of the additive joint entropy, and almost all of its compressible margin is on the exponent. Codec designers should therefore concentrate effort on exponent prediction and on cross-stream coders that bring information from outside the value distribution.

11 Discussion

The three-component decomposition is structurally trivial; the BF16 codepoint is defined as the triple (s, e, m) . What is not trivial is that the empirical mutual information between exponent and mantissa under trained-weight distributions is so small. We report this as a measured property of the tested populations and not as a mathematical theorem; nothing about the BF16 format itself forces $MI(e; m)$ to be small. A poorly trained or untrained network with structured exponent-mantissa correlations would give a larger value, and the additive decomposition would correspondingly become a less good approximation.

12 Limitations

We note four limitations.

1. The near-independence result is an empirical statement for the tested trained-weight populations. We do not claim universality beyond those networks; in particular, the result should not be assumed for newly initialised, partially trained, or pathologically structured networks.
2. The reported entropies are reported to three decimal places. The third decimal place is sensitive to the Miller bias correction and to the choice of population subset.
3. The decomposition is into the format-defined components (s, e, m) . Alternative decompositions, for example into pseudo-categories defined over multi-bit substrings, may yield smaller joint entropies; they are out of scope here.
4. Public reproducibility is limited by disclosure control. Internal verification artefacts, including the per-component frequency tables and the joint exponent-mantissa table, are maintained by Forlais Group but are not released with this paper.

13 Data and code availability

Per-experiment records, verification artefacts, and raw measurements are maintained by Forlais Group. Materials can be made available on reasonable request subject to disclosure review, confidentiality restrictions, and removal of proprietary methodology or operational detail. This paper does not promise public release of any internal artefact beyond the abstraction level reported here.

14 Competing interests

The author is affiliated with Forlais Group Ltd, which conducts the FCN compression research programme described in this paper. The author declares no other competing financial or non-financial interests.

15 Conclusion

The FCN evidence supports a scoped near-independence statement: stored BF16 weight values, decomposed into sign, exponent, and mantissa under the BF16 layout, satisfy the additive entropy identity to within an off-diagonal mutual-information term of approximately 0.047 bits per value across the tested trained-weight populations. The per-component entropies are approximately 1.000, 2.611, and 6.972 bits respectively, giving a joint entropy of approximately 10.536 bits per value and a compression ceiling of approximately $1.518\times$ at that joint entropy. The result licenses per-component codec design at single-model entropy scope and identifies the exponent as the locus of compressible margin under the tested conditions. The result is reported here in its public-safe form and should be read alongside the limitations enumerated above.

References

- [1] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2 edition, 2006.
- [2] Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems*, 2022.

- [3] Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations*, 2023.
- [4] Yefan Han, Yidan Wang, Yiding Jiang, and Hamed Pirsiavash. DFloat11: Lossless compression of large language models via dynamic-length floating-point, 2025. Independent published method on lossless weight compression.
- [5] IEEE. IEEE standard for floating-point arithmetic. IEEE Std 754-2019 (Revision of IEEE 754-2008), 2019.
- [6] Dhiraj Kalamkar, Dheevatsa Mudigere, Naveen Mellempudi, Dipankar Das, Kunal Banerjee, Sasikanth Avancha, Dharma Teja Vooturi, Nataraj Jammalamadaka, Jianyu Huang, Hector Yuen, Jiyan Yang, Jongsoo Park, Alexander Heinecke, Evangelos Georganas, Sudarshan Srinivasan, Abhisek Kundu, Misha Smelyanskiy, Bharat Kaul, and Pradeep Dubey. A study of BFLOAT16 for deep learning training, 2019.
- [7] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, and Hao Wu. Mixed precision training. In *International Conference on Learning Representations*, 2018.
- [8] George A. Miller. Note on the bias of information estimates. *Information Theory in Psychology*, pages 95-100, 1955.
- [9] Liam Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6):1191-1253, 2003.
- [10] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379-423, 1948.

Revision History

Version	Date	Change summary	Owner
0.1	14 May 2026	Initial FCN academic paper package generated from source evidence.	Oliver Cooper
0.2	14 May 2026	Expanded references, structured evidence summary, scope, and limitations.	Oliver Cooper
0.3	14 May 2026	Full evidence-grounded manuscript with public-safe abstraction, table summaries, and category-aware framing.	Oliver Cooper
1.0	14 May 2026	Publication-candidate manuscript: additive-decomposition analysis, cross-architecture consistency, and implications for codec design completed.	Oliver Cooper